

Donchian + ADX Strategy

Breakout Trading with Trend Strength Confirmation

Detailed Strategy Guide -- SwingPortfolio

1. What is the Donchian Channel?

The Donchian Channel is a volatility-based indicator developed by Richard Donchian, widely regarded as the father of trend-following. It plots two bands around price: the highest high and the lowest low over a lookback period (typically 20 days).

The logic is simple: when price breaks above the 20-day high, it suggests an uptrend has started. When it breaks below the 20-day low, it suggests a downtrend. Donchian used this as the basis for his famous Turtle Trading System.

```
Donchian Upper = Highest High (last N periods)
Donchian Lower = Lowest Low (last N periods)
where N = donchian_period (default: 20)
```

In our implementation:

- dc20 = rolling(20).max() of daily highs
- dc20_low = rolling(20).min() of daily lows

2. What is ADX?

ADX (Average Directional Index) was developed by J. Welles Wilder to measure trend strength. Unlike many oscillators, ADX does NOT tell you direction -- it only tells you how strong the trend is. It ranges from 0 to 100, with values above 25 typically indicating a strong trend.

ADX is derived from two directional indicators:

- +DI (Positive Directional Indicator): Measures upward price movement strength
- -DI (Negative Directional Indicator): Measures downward price movement strength

When +DI > -DI, the bulls are in control (uptrend). When -DI > +DI, the bears are in control (downtrend). The crossover of +DI and -DI is itself a trading signal.

ADX Calculation (14-period default)

Step 1: True Range (TR)

$$TR = \max(High - Low, |High - prevClose|, |Low - prevClose|)$$

Step 2: Directional Movement (+DM, -DM)

$$+DM = High - prevHigh \text{ (if } > prevLow - Low \text{ and } > 0, \text{ else } 0)$$

$$-DM = prevLow - Low \text{ (if } > High - prevHigh \text{ and } > 0, \text{ else } 0)$$

Step 3: Smooth +DM and -DM (Wilder's EMA, alpha = 1/14)

$$+DI = 100 * \text{smoothed}(+DM) / \text{smoothed}(TR)$$

$$-DI = 100 * \text{smoothed}(-DM) / \text{smoothed}(TR)$$

Step 4: Directional Index (DX)

$$DX = 100 * |+DI - -DI| / (+DI + -DI)$$

Step 5: ADX = smoothed(DX) over 14 periods

Note: ADX lags. A value above 25 means the trend has been in place for several days. It is a CONFIRMATION tool, not a prediction tool.

3. Entry Conditions -- Donchian + ADX

LONG Entry (Bullish)

- Condition 1: Close > Previous day's Donchian Upper (dc20)
- Meaning: Price has broken above the 20-day high -- a breakout
- Condition 2: ADX > 25
- Meaning: The breakout is happening during a strong trend, not a fakeout
- Condition 3: +DI > -DI
- Meaning: The uptrend is confirmed -- bulls are in control

SHORT Entry (Bearish)

- Condition 1: Close < Previous day's Donchian Lower (dc20_low)
- Meaning: Price has broken below the 20-day low
- Condition 2: ADX > 25
- Condition 3: -DI > +DI
- Meaning: The downtrend is confirmed -- bears are in control

Why Previous Day's Donchian?

The code uses `prev['dc20']` and `prev['dc20_low']` rather than today's. This is important because the breakout is confirmed on today's close, but the breakout level itself was established yesterday. Using yesterday's level ensures we're catching a CLOSE ABOVE a level, not a level that moved with today's price.

4. Exit Conditions

Stop-Loss (SL)

The stop-loss for Donchian ADX uses the ATR (Average True Range) to set a volatility-adaptive stop. The distance from entry is:

```
LONG: SL = reference_close - (ATR x SL_ATR_MULTIPLIER)
SHORT: SL = reference_close + (ATR x SL_ATR_MULTIPLIER)
```

Where SL_ATR_MULTIPLIER = 3.0 (current setting)
And reference_close = the close price on the signal day

With ATR multiplier = 3, the stop is placed at 3x the stock's average daily range from the reference price. This gives a swing trade room to breathe -- about 1.5x the daily range beyond the entry.

Trailing Stop-Loss

Once the trade moves in our favor, the stop-loss trails behind:

- LONG: Trail = max(SL, BestPrice x (1 - TRAIL_PCT))
- SHORT: Trail = min(SL, BestPrice x (1 + TRAIL_PCT))

TRAIL_PCT = 3% (current setting). This means once price moves 3% in our favor, the stop moves to lock in 3% of the gain. As price continues favorably, the trail continues to lock in profits.

Target

The target is set at a fixed 1:1.5 risk-reward ratio:

```
LONG: Target = Entry + (ATR x SL_ATR_MULTIPLIER x RR_RATIO)
SHORT: Target = Entry - (ATR x SL_ATR_MULTIPLIER x RR_RATIO)
```

With RR_RATIO = 1.5: risk 3x ATR to gain 4.5x ATR

Hard SL (Safety Net)

A hard stop of Rs. 75,000 per position acts as a backstop. If the ATR-based SL is wider than Rs. 75K per lot, the hard stop overrides it. This prevents catastrophic losses during extreme volatility.

End of Session

At 15:15, any remaining OPEN positions are NOT closed -- they carry forward to the next trading day. This is true swing behavior. The only exception is the last day of the backtest month where all remaining positions are closed.

5. Worked Example -- BAJAJFINSV SHORT, June 8-9 2026

Let us trace through a real trade from our backtest:

Signal Day: June 8 (Monday) -- Using June 5 (Friday) close data

Date	Open	High	Low	Close	ADX	+DI
05-Jun	1684.00	1696.00	1662.00	1674.60	27.7	18.5

On June 5 (Friday's close):

- Close = 1674.60
- 20-day high (dc20) = 1696.80 (highest high in last 20 days)
- 20-day low (dc20_low) = 1662.00 (lowest low in last 20 days)
- ADX = 27.7 (> 25 -- strong trend)
- +DI = 18.5, -DI = 22.3 (-DI > +DI -- bears in control)

Signal check: Close (1674.60) < prev dc20_low (1681.20)?

YES -- SHORT signal triggered.

Entry Day: June 8 (Monday)

The backtest enters on the signal day's open/intraday via zscore method. For this date, zscore entry triggered at 09:48 at 231.38.

SL Calculation:

```
Reference close: 1674.60 (Friday's close)
ATR(14): 36.24
SL = 1674.60 + (36.24 x 3) = 1674.60 + 108.72 = 1783.32
Entry price (via zscore): 1676.30
Hard SL per share: 75,000 / 300 = 250.00
Hard SL price: 1676.30 + 250.00 = 1926.30
Final trail = min(1783.32, 1926.30) = 1783.32 (ATR SL wins)
```

Monitoring Days

June 8: Low = 1673.10. SL not hit (1783.32). Position stays open.

June 9: High = 1696.20. SL not hit. At EOD, position carried forward.

...

Exit: Eventually exited at 1693.50 (EOD close) for P&L = -5,160.

6. Parameter Tuning Guide

donchian_period (default: 20)

- Range: 10-50
- Lower values (10-15): More signals, more false breakouts
- Higher values (30-50): Fewer signals, stronger trends only
- For swing trading (multi-day): 20 is a good balance

adx_threshold (default: 25)

- Range: 20-35
- Lower (20): More trades but includes weak trends
- Higher (30-35): Only trades in very strong trends -- fewer but higher quality
- Wilder's original: 25 is the standard cutoff

SL_ATR_MULTIPLIER (default: 3.0)

- Range: 1.5-5.0
- Lower (1.5-2.0): Tight stops -- exits quickly, higher win rate but smaller profits
- Higher (3.0-5.0): Wide stops -- gives room, but larger losses when wrong
- Our setting (3.0): 1.5x the daily range beyond entry -- balanced

TRAIL_PCT (default: 0.03)

- Range: 0.01-0.05
- Lower (0.01): Tight trail -- locks profits fast but exits on pullbacks
- Higher (0.03-0.05): Loose trail -- holds through pullbacks, lets runners run
- Was 0.0075 before -- caused premature exits. 3% is appropriate for swing.

RR_RATIO (default: 1.5)

- Range: 1.0-3.0
- Higher values (2-3): Fewer targets hit but when hit, profits are large
- Lower values (1-1.5): More targets hit, but smaller profits per trade
- 1.5 is a reasonable compromise for swing trading

7. Common Pitfalls & How to Avoid Them

Pitfall 1: Trading Ranging Markets

Donchian + ADX works best in trending markets. In sideways/choppy markets, breakouts fail frequently (false breakouts). $ADX < 25$ is the market telling you there is no trend -- respect it and stay out.

Pitfall 2: Ignoring DI Crossovers

The strategy requires both $ADX > 25$ AND DI confirmation. Even if ADX is high, if +DI and -DI are close together, the direction is unclear. Wait for a clear crossover.

Pitfall 3: Stop-Loss Too Tight (Original Setting)

Our original SL of 2x ATR was being hit on normal intraday volatility. The increase to 3x ATR + 3% trail gives swing trades room to survive pullbacks. Check the AvgHold of 4 days -- we want trades lasting multiple days.

Pitfall 4: Overtrading

Donchian can generate many breakouts in volatile periods. With max 2 positions per strategy and ADX ranking (highest ADX first), we ensure only the strongest breakouts are taken.

8. How Donchian ADX Fits in the Portfolio

Donchian ADX complements Keltner RSI because it catches different market conditions:

- Donchian excels in: Strong trending markets with clear breakouts
- Donchian struggles in: Low-volatility, ranging, or mean-reverting markets
- Keltner excels in: Momentum continuation with volume confirmation
- Keltner struggles in: Low-volume breakouts or news-driven spikes

When one strategy is having a losing week, the other is often profitable. In 30 weeks of backtesting, only 1 week saw BOTH strategies lose simultaneously.

Key metrics summary:

70 trades over 7 months

69% win rate

+721,364 P&L (max 2 positions)

Max drawdown: Rs. 66K (13% of capital)

Average hold: 4 days